Lecture 4. Exploratory Data Analysis

Variation, missing values, covariation, patterns and models

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4.1 What EDA is

Motivation. The tools of Lectures 2 and 3 are now combined for their real purpose: understanding a data set before any formal analysis. EDA has no fixed procedure; it is a habit of asking questions and letting each answer suggest the next one.

The goal of **exploratory data analysis** (EDA) is to develop an understanding of your data. It is an iterative cycle:

1. generate questions about the data;
2. search for answers by visualising, transforming and modelling;
3. use what you learn to refine the questions or generate new ones.

Early on, investigate every idea; some will pan out and some will be dead ends. EDA also includes **data cleaning**: checking whether the data meet your expectations needs the same tools.

Two quotations set the tone: "There are no routine statistical questions, only questionable statistical routines" (D. R. Cox), and "Far better an approximate answer to the right question, which is often vague, than an exact answer to the wrong question, which can always be made precise" (J. W. Tukey). Two questions are always useful:

- What type of **variation** occurs *within* my variables?
- What type of **covariation** occurs *between* my variables?

Definitions. A *variable* is a quantity, quality or property you can measure; a *value* is its state when measured; an *observation* is a set of measurements made under similar conditions (usually on the same object at the same time), containing several values; *tabular data* is a set of values, each associated with a variable and an observation.

4.2 Variation

Motivation. Every variable varies, from measurement error or from real differences between objects, and the pattern of that variation is itself information. The way to see it is to plot the variable's distribution.

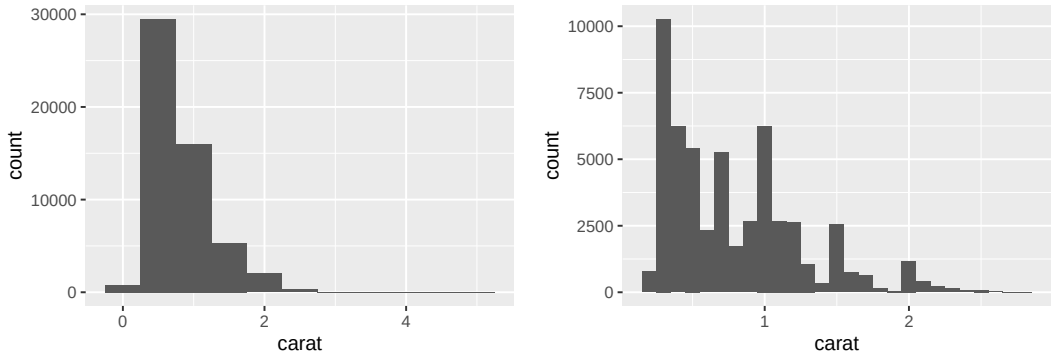
Variation is the tendency of the values of a variable to change from measurement to measurement. How to visualise a distribution depends on the type of variable.

Categorical variables take a small set of values; use a bar chart, whose heights are the counts:

```
ggplot(data = diamonds) + geom_bar(mapping = aes(x = cut))
diamonds %>% count(cut)                    # the same numbers
```

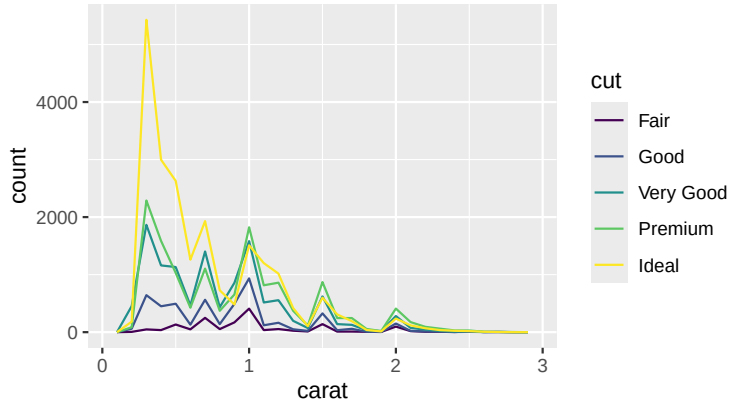
Continuous variables (numbers, date-times) take infinitely many ordered values; use a histogram, which counts the observations in equal-width bins:

```
ggplot(data = diamonds) +
  geom_histogram(mapping = aes(x = carat), binwidth = 0.5)
smaller <- diamonds %>% filter(carat < 3)
ggplot(data = smaller, mapping = aes(x = carat)) +
  geom_histogram(binwidth = 0.1)
```



`diamonds %>% count(cut_width(carat, 0.5))` computes the bars by hand. Always try several binwidths: different binwidths reveal different patterns. To overlay several distributions, `geom_freqpoly()` draws the counts as lines, which overlap better than bars:

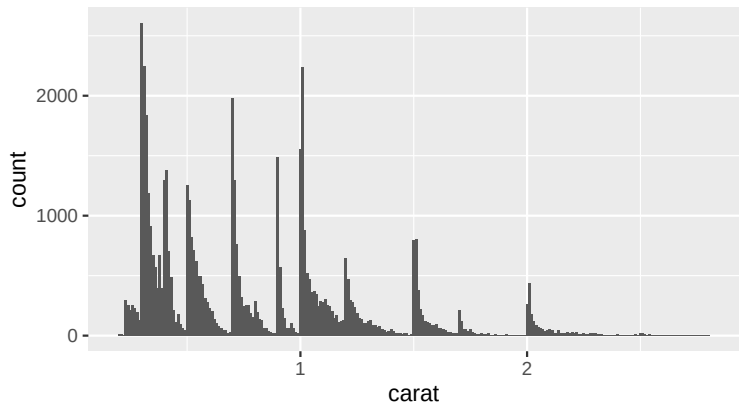
```
ggplot(data = smaller, mapping = aes(x = carat, colour = cut)) +
  geom_freqpoly(binwidth = 0.1)
```



4.2.1 Typical values

Turn the picture into questions: Which values are most common, and why? Which are rare, and does that match expectations? Are there unusual patterns? With a fine binwidth:

```
ggplot(data = smaller, mapping = aes(x = carat)) +
  geom_histogram(binwidth = 0.01)
```



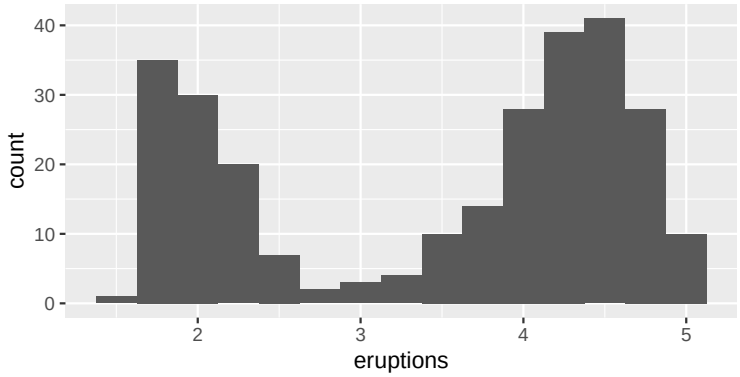
Why are there more diamonds at whole carats and common fractions? Why are there more diamonds

slightly to the right of each peak than to the left? Why none above 3 carats? (A plausible answer to the first two: cutters aim for round weights because buyers pay a premium for them [Supplement].)

4.2.2 Subgroups

Clusters of similar values suggest subgroups. Ask how observations within a cluster are similar, how clusters differ, how to explain them, and whether the clusters could be misleading. The eruption lengths of the Old Faithful geyser fall into short (about 2 minutes) and long (4–5 minutes) eruptions with little in between:

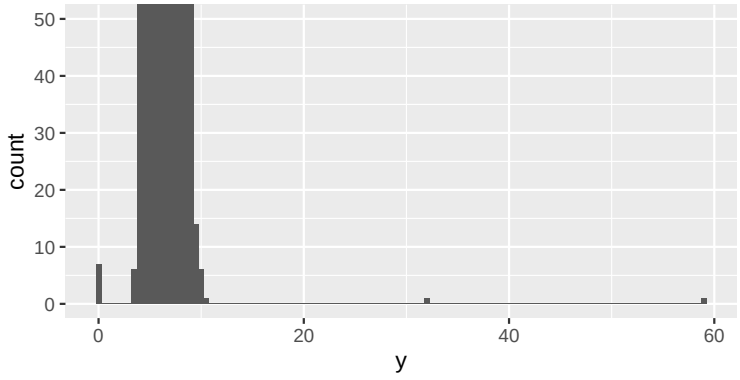
```
ggplot(data = faithful, mapping = aes(x = eruptions)) +
  geom_histogram(binwidth = 0.25)
```



4.2.3 Unusual values

The histogram of the width `y` of diamonds has a suspiciously wide x-axis. The rare values are there but their bars are too short to see; zooming with `coord_cartesian(ylim = c(0, 50))` reveals values near 0, 30 and 60 mm. (`coord_cartesian()` zooms without dropping data; `ylim()` would drop it.)

```
ggplot(diamonds) +
  geom_histogram(mapping = aes(x = y), binwidth = 0.5) +
  coord_cartesian(ylim = c(0, 50))
```




```
unusual <- diamonds %>%
  filter(y < 3 | y > 20) %>%
  select(price, x, y, z) %>%
  arrange(y)
unusual
```

```
## # A tibble: 9 x 4
##   price     x     y     z
##   <int> <dbl> <dbl> <dbl>
## 1  5139     0     0     0
## 2  6381     0     0     0
## 3 12800     0     0     0
## 4 15686     0     0     0
## 5 18034     0     0     0
## # i 4 more rows
```

A width of 0 mm is impossible, so those are errors (here all of `x`, `y`, `z` are 0 for some). Widths of 32 and 59 mm are implausible: such diamonds would be over an inch wide yet are priced like ordinary stones. Outliers are sometimes data-entry errors and sometimes important new science. Good practice:

- repeat the analysis with and without the outliers;
- if they have little effect and you cannot explain them, replace them with missing values and move on;
- if they have a substantial effect, do not drop them without justification: find the cause and disclose the removal.

4.3 Missing values

Motivation. After finding bad values we must decide what to do with them. Deleting whole rows throws away the good measurements in those rows too; marking only the bad value as missing keeps them.

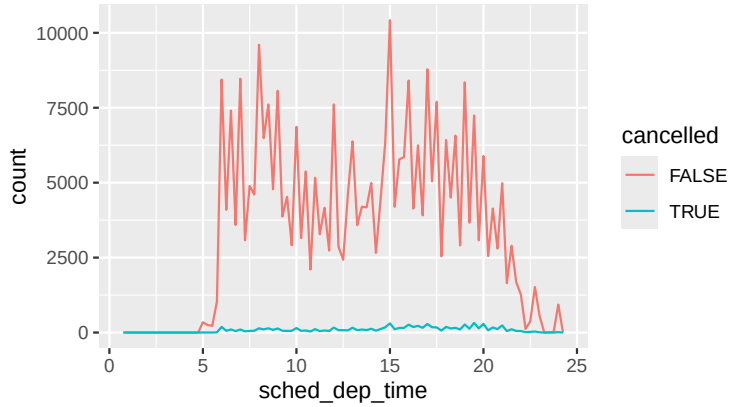
Rather than dropping rows (`filter(between(y, 3, 20))`), replace unusual values by `NA`:

```
diamonds2 <- diamonds %>%
  mutate(y = ifelse(y < 3 | y > 20, NA, y))
```

`ifelse(test, yes, no)` is vectorised: where the logical vector `test` is `TRUE` it returns the value of `yes`, elsewhere the value of `no`. (`case_when()` handles more than two cases.) ggplot2 does not plot missing values but warns that they were removed; `geom_point(na.rm = TRUE)` suppresses the warning.

Sometimes missingness itself is informative. In `flights`, a missing `dep_time` means a cancelled flight, so we can compare scheduled departure times of cancelled and non-cancelled flights:

```
flights %>%
  mutate(
    cancelled = is.na(dep_time),
    sched_hour = sched_dep_time %/% 100,
    sched_min = sched_dep_time %% 100,
    sched_dep_time = sched_hour + sched_min / 60
  ) %>%
  ggplot(mapping = aes(sched_dep_time)) +
  geom_freqpoly(mapping = aes(colour = cancelled), binwidth = 1/4)
```



The comparison is poor because cancelled flights are few; the next section shows how to compare groups of very different sizes.

4.4 Covariation

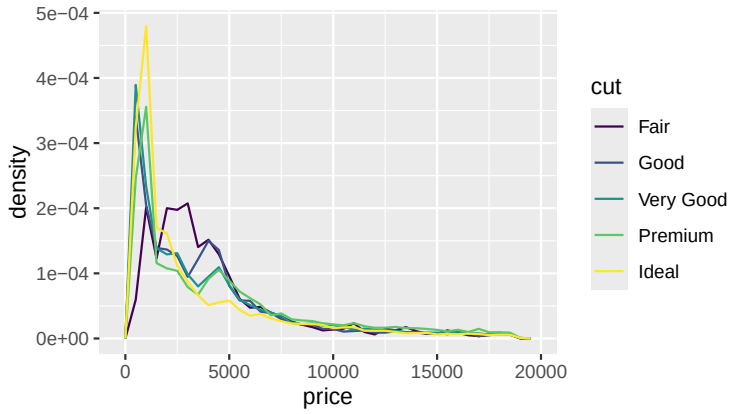
Motivation. Variation describes one variable; covariation describes how two vary together, which is where most interesting questions live. The right plot depends on the types of the two variables.

Covariation is the tendency of two or more variables to vary together in a related way. The best way to spot it is to visualise the relationship.

4.4.1 A categorical and a continuous variable

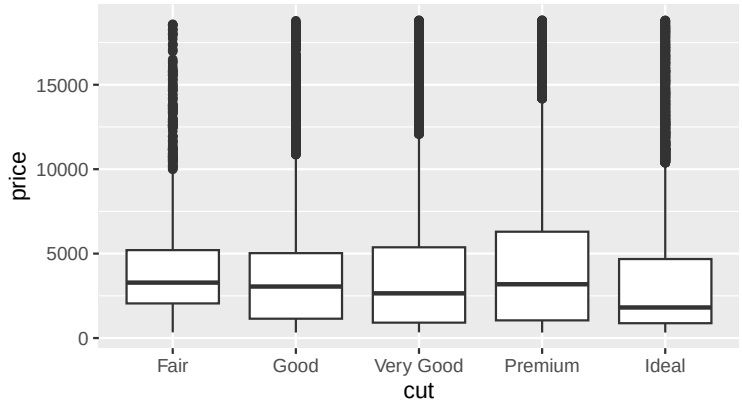
Frequency polygons of price by cut are hard to compare because the counts differ so much between cuts. Displaying **density** instead, the count standardised so that each polygon has area one, makes the shapes comparable:

```
ggplot(data = diamonds, mapping = aes(x = price, y = after_stat(density))) +
  geom_freqpoly(mapping = aes(colour = cut), binwidth = 500)
```



Surprisingly, fair diamonds (the lowest quality) seem to have the highest average price. A **boxplot** summarises each distribution compactly: the box spans the interquartile range (IQR) with a line at the median, whiskers extend to the farthest non-outlying points, and points more than 1.5×IQR beyond the box are drawn individually.

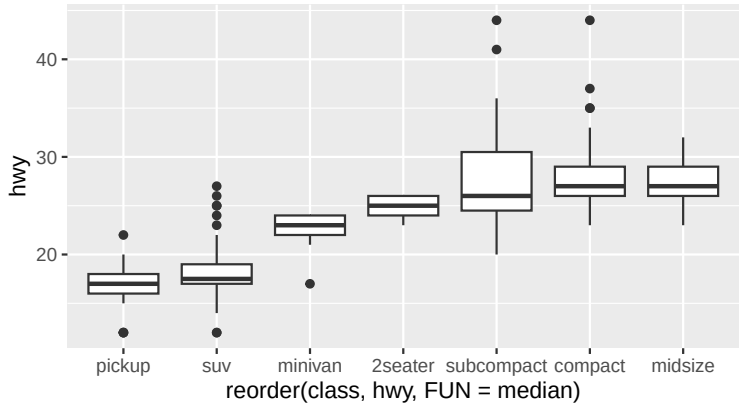
```
ggplot(data = diamonds, mapping = aes(x = cut, y = price)) +
  geom_boxplot()
```



It confirms the counterintuitive finding: better diamonds are cheaper on average. (Section 4.5 explains why.)

Categories without an intrinsic order are easier to read when reordered by a summary of the continuous variable:

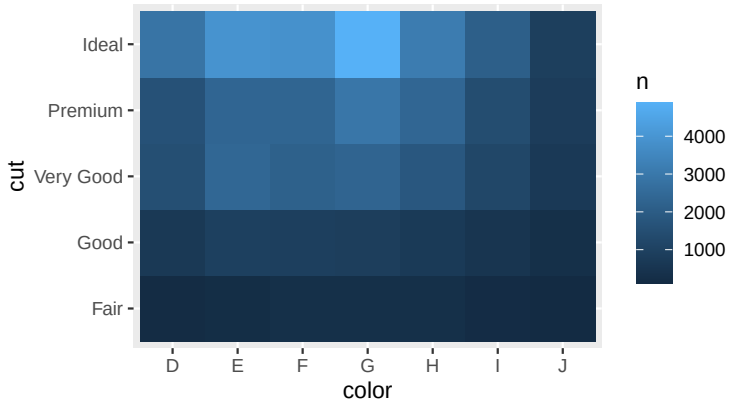
```
ggplot(data = mpg) +
  geom_boxplot(mapping = aes(x = reorder(class, hwy, FUN = median), y = hwy))
```



4.4.2 Two categorical variables

Count the observations for each combination, either with `geom_count()` (circle size proportional to count) or by computing the counts with dplyr and drawing a heat map:

```
diamonds %>%
  count(color, cut) %>%
  ggplot(mapping = aes(x = color, y = cut)) +
  geom_tile(mapping = aes(fill = n))
```

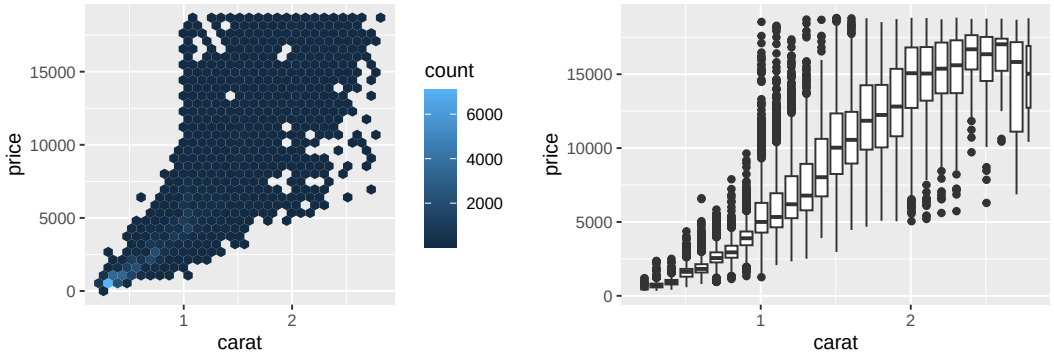


4.4.3 Two continuous variables

A scatterplot is the natural choice, but with 54 000 points it overplots. Remedies: transparency (`alpha = 1/100`), or binning in two dimensions with `geom_bin2d()` (rectangles) or `geom_hex()` (hexagons, needs `hexbin`), which colour each bin by its count. Another option bins one variable so it acts like a categorical one:

```
ggplot(data = smaller) +
  geom_hex(mapping = aes(x = carat, y = price))
ggplot(data = smaller, mapping = aes(x = carat, y = price)) +
  geom_boxplot(mapping = aes(group = cut_width(carat, 0.1)))
```

```
## Warning: Orientation is not uniquely specified when both the x and y aesthetics are continuous.
## Picking default orientation 'x'.
```



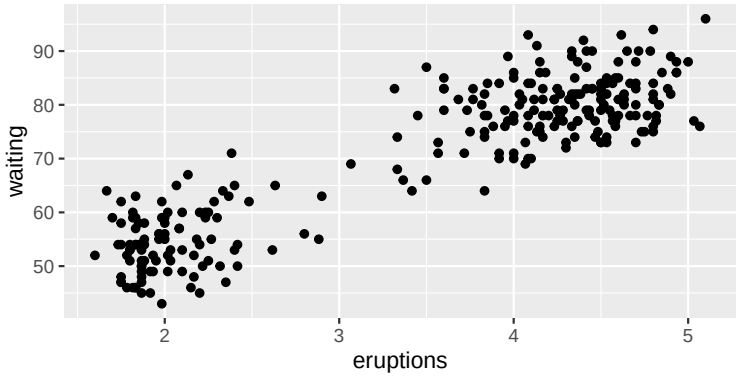
`cut_width(x, w)` makes bins of width `w`; `cut_number(x, n)` makes `n` bins with approximately the same number of points each.

4.5 Patterns and models

Motivation. A pattern suggests a relationship, but a strong relationship can hide weaker ones. A model lets us remove the strong one and look at what is left, which resolves the puzzle of the cheap good diamonds.

When a systematic relationship exists between two variables, it appears as a pattern. Then ask: could it be coincidence? How can the relationship be described, and how strong is it? What other variables might affect it? Does it change within subgroups? The Old Faithful scatterplot shows that longer waiting times go with longer eruptions, and also shows the two clusters:

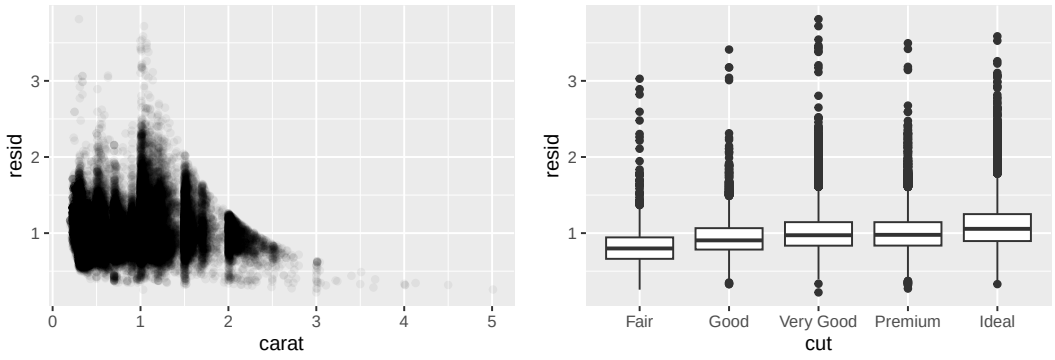
```
ggplot(data = faithful) +
  geom_point(mapping = aes(x = eruptions, y = waiting))
```



Patterns reveal covariation. Variation creates uncertainty; covariation reduces it, because knowing one variable improves prediction of the other. Only if the covariation is causal can one variable be used to *control* the other.

Models extract patterns. In `diamonds`, price and carat are tightly related, and cut is related to carat (better cuts tend to be smaller stones). To see the effect of cut alone, fit a model of price on carat and look at the residuals, the part of price the model does not explain:

```
mod <- lm(log(price) ~ log(carat), data = diamonds)
diamonds2 <- diamonds %>%
  add_residuals(mod) %>%
  mutate(resid = exp(resid))
ggplot(data = diamonds2) +
  geom_point(mapping = aes(x = carat, y = resid), alpha = 1/20)
ggplot(data = diamonds2) +
  geom_boxplot(mapping = aes(x = cut, y = resid))
```



Once the strong price–carat relationship is removed, the expected order appears: *relative to their size*, better diamonds are more expensive. Lectures 13 and 14 study models in detail.

Why log–log [Supplement]

The model $\log(\text{price}) = a_0 + a_1 \log(\text{carat})$ says $\text{price} = e^{a_0}\,\text{carat}^{a_1}$, a power law. On the log scale the relationship is linear with roughly constant spread, which a straight line fits well. The residual on the log scale is $\log(\text{price}/\widehat{\text{price}})$, so `exp(resid)` is the ratio of the actual price to the price predicted from carat alone: values above 1 mean more expensive than a typical diamond of that size.

```r
coef(mod)
```

```
## (Intercept)  log(carat)
##    8.448661    1.675817
```